9 Figure 9.1 shows a beam of α-particles and γ-radiation entering an electric field between two charged metal plates P and Q.

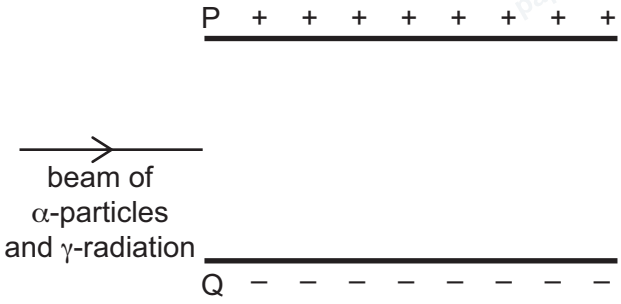

Figure 9.1

(a) On Figure 9.1, draw the path of:

1 the α-particles inside the electric field. Label this line α.

2 the γ-radiation inside the electric field. Label this line γ.

[3]

(b) A student places a sheet of paper in the path of the beam before it reaches the electric field. State how this affects the path of the α-particles and γ-radiation inside the electric field.

α-particles .....

γ-radiation .....

[2]

(c) The beam of α-particles and γ-radiation in Figure 9.1 is replaced by a beam of β-particles.

State and explain the path of the β-particles in the electric field. Use ideas about electrostatic force in your answer.

statement .....

explanation .....

.....

[3]

[Total: 8]

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